

Operations Research: theory and applications to networking

Graph coloring revisited

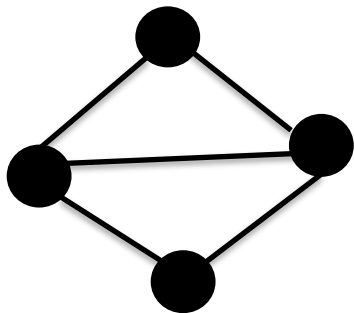
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ICT4SS-CE

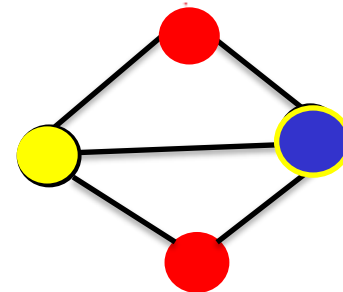
Graph Coloring problem

Given an undirected graph, assign colors (from a palette) to nodes so that: different colors are assigned to adjacent nodes.

The goal is to minimize the number of used colors.



Formulate the problem!



Upper bound to $\chi(\mathcal{G})$

Provide a simple, non trivial upper bound to the chromatic number $\chi(\mathcal{G})$ (minimum number of colors needed).

Hint: compare $\chi(\mathcal{G})$ with the maximum degree $\Delta(\mathcal{G})$

Particular graphs

Find the chromatic number $\chi(G)$ in:

- Trees,
- Bipartite graphs,
- Cicles,
- Cliques.

Lower bound to $\chi(\mathcal{G})$

Find a non trivial lower bound to $\chi(\mathcal{G})$.

Graph coloring

As for graph coloring consider graphs with $N = 15, 30, 60$, in which for every pair of nodes (i,j) , an undirected edge (i,j) is added to the graph with probability p , independently from other edges.

For $p = 0.2$ and 0.4 generate three/four instances and solve the GC problem on them.